



# **Building India's Sovereign AI Stack: From Tokens to Foundational Scale Large Language Models (LLMs) to Intelligent Agents**

**Piyush Sawarkar**

*Founding Member & Tech Lead Text Foundation Models @ BharatGen, India's Sovereign AI National Mission   
IIT Bombay Alumnus*

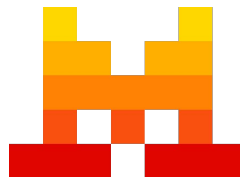
---

Advances in Proteomics Technologies Conference, 2026  
IIT Bombay • June 29, 2026

# What is Sovereign AI?



sakana.ai



**Mistral AI**

# The Case for Sovereign AI

01

## Sovereign

---

A guarantee that AI access cannot be removed by another country's politics or a company's boardroom.

*Supply security.*

*Provenance.*

*Serviceability.*

02

## Accessible

---

Cost is a design feature.

AI that reaches an Indian consumer, a farmer, a small business.

*Inclusive.*

*Voice-first.*

*Inference-light.*

03

## Indian by design

---

AI that carries Indian instinct. Rooted in language, culture, and how a billion people think and speak.

*22 languages.*

*Code-mixed speech.*

*Indian perspective.*

# The Case for Sovereign AI



## India's first government-funded sovereign AI initiative

A not-for-profit organisation with a network of India's top academic institutes, anchored at IIT Bombay  
Backed by DST and the IndiaAI Mission, MEITY



### Sovereign AI Ecosystem

National AI infrastructure owned by India. Supply security, provenance, and accountability



### Foundational Models

Purpose-built models for text, speech, and documents across 22 Indian languages



### Research

SOTA optimisation and algorithms. 915+ research papers from 50+ global experts across 9 of India's top institutions



### Startups & Deployments

Knowledge-driven inclusivity. Real-world deployments across BFSI, healthcare, governance, and education

# BharatGen Launched Flagship Models at IndiaAI Summit, Feb'26

WHAT WE'VE BUILT



## A multimodal, multilingual foundational AI stack

TEXT · FLAGSHIP MODEL

### Param2

Reasoning · Code · Math · Tool use

17 billion Mixture-of-Expert

2.4 billion active tokens

Trained on 22 trillion tokens

22 Indian languages + English

Multilingual input

MoE Routing

Reasoning, code,  
language

SPEECH · ASR

### Shrutam2

LLM-powered · MoE projector · 12 Indian languages

SPEECH · TTS

### Sooktam2

Voice cloning across languages · CLS · 11 Indian languages + English

VISION · DOCUMENT VLM

### Patram

Vision transformer · Document QA · Multi-domain document support

# What it takes to build a Foundational AI from Scratch?



# The Tokenizer

Many words map to one token, but some don't: indivisible.

Unicode characters like emojis may be split into many tokens containing the underlying bytes: 🖐

Sequences of characters commonly found next to each other may be grouped together: 1234567890



After Tokenization

Many words map to one token, but some don't: indivisible.

Unicode characters like emojis may be split into many tokens containing the underlying bytes: 🖐

Sequences of characters commonly found next to each other may be grouped together: 1234567890

# The Tokenizer

A tokenizer converts raw text into numbers the model can process. It breaks words into chunks called tokens. The model never sees the original text, it only sees sequences of numbers representing tokens.

## What is a token?

A token is not necessarily a word. It is the basic unit the model is trained on. OpenAI's tokenizer, for example, splits "Bombay" into two tokens: "Bomb" + 'ay". BharatGen's tokenizer treats "Bombay" as one token, because it was trained on Indian data and "Bombay" is a well-known place.

| Word                      | OpenAI tokenizer                                | BharatGen tokenizer                    |
|---------------------------|-------------------------------------------------|----------------------------------------|
| Bombay                    | 2 tokens: "Bomb" + 'ay" → charged 2x            | 1 token: "Bombay" → charged 1x         |
| Hindi word (e.g., नमस्ते) | May split into 6–7 tokens (byte-level fallback) | 1–2 tokens (trained on Indian scripts) |

# The Tokenizer

## Fertility: the key metric

Fertility is the ratio of tokens to words. Lower is better: it means the tokenizer is more efficient at representing the language.

|                                                                           |                                                                                           |                                                                                       |
|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Fertility (English)<br><b>~1.4</b><br>tokens per word (industry standard) | Fertility (Indian langs, global models)<br><b>6–7x</b><br>tokens per Indian-language word | Fertility (Indian langs, BharatGen)<br><b>1–3x</b><br>tokens per Indian-language word |
|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|

Why fertility matters commercially: API pricing is charged per token. If a global model charges \$10 per 1M output tokens and uses 6–7 tokens for every Indian-language word, an Indian enterprise using it in Hindi is paying 4–5x more than they would with BharatGen's model for the same output. This is a direct cost argument when talking to clients.

# BharatGen's Tokenizer Fertility

| Language  | AdaptMix<br>(Param) | Qwen        | LLaMA       | Nemotron<br>Mistral | Nemotron<br>Mini | Sarvam-M    | DeepSeekv3  | Gemma 27B   | Airavata    |
|-----------|---------------------|-------------|-------------|---------------------|------------------|-------------|-------------|-------------|-------------|
| Assamese  | <b>1.93</b>         | 7.18        | 8.06        | 4.24                | 4.58             | 4.24        | 3.59        | 2.68        | 8.94        |
| Bengali   | 1.90                | 6.92        | 7.85        | 2.93                | 2.65             | 2.93        | 2.89        | <b>1.74</b> | 8.20        |
| English   | 1.47                | 1.36        | 1.35        | 1.37                | 1.35             | 1.37        | 1.33        | <b>1.35</b> | 1.58        |
| Gujarati  | <b>2.03</b>         | 8.53        | 9.54        | 3.59                | 15.17            | 3.59        | 4.84        | 2.39        | 14.14       |
| Hindi     | 1.43                | 4.66        | 2.65        | 1.97                | 1.77             | 1.97        | 2.92        | <b>1.38</b> | 1.80        |
| Kannada   | <b>2.30</b>         | 11.08       | 13.81       | 3.82                | 4.02             | 3.82        | 5.83        | 3.15        | 19.35       |
| Maithili  | <b>1.73</b>         | 4.67        | 2.85        | 2.53                | 2.28             | 2.53        | 3.28        | 1.90        | 2.45        |
| Malayalam | <b>2.60</b>         | 13.30       | 16.00       | 4.88                | 4.71             | 4.88        | 7.83        | 3.39        | 11.38       |
| Marathi   | <b>1.87</b>         | 6.46        | 3.86        | 3.14                | 2.62             | 3.14        | 4.15        | 1.94        | 3.31        |
| Nepali    | <b>1.70</b>         | 6.28        | 3.61        | 3.04                | 2.32             | 3.04        | 4.07        | 2.06        | 3.05        |
| Oriya     | 1.95                | 12.92       | 15.91       | <b>1.71</b>         | 17.24            | 17.23       | 7.26        | 4.60        | 17.29       |
| Punjabi   | <b>1.61</b>         | 7.39        | 7.88        | 3.12                | 12.70            | 3.12        | 4.51        | 2.74        | 10.84       |
| Sanskrit  | <b>2.57</b>         | 8.00        | 4.75        | 4.26                | 4.32             | 4.26        | 4.95        | 3.36        | 4.66        |
| Sindhi    | <b>1.67</b>         | 3.09        | 2.99        | 2.65                | 2.83             | 2.65        | 2.98        | 2.14        | 5.04        |
| Tamil     | <b>2.35</b>         | 9.75        | 11.89       | 3.71                | 3.57             | 3.71        | 4.88        | 2.42        | 10.50       |
| Telugu    | <b>2.34</b>         | 11.45       | 13.30       | 3.90                | 3.77             | 3.90        | 5.99        | 2.93        | 19.51       |
| Average   | <b>1.97</b>         | <b>7.69</b> | <b>7.89</b> | <b>3.18</b>         | <b>5.37</b>      | <b>4.15</b> | <b>4.46</b> | <b>2.51</b> | <b>8.88</b> |

# The Model Development Lifecycle (MDLC)

## ► HIGH-LEVEL TRAINING PIPELINE

### Raw Intelligence

just pre-trained  
next-token prediction



### Instruction-following Intelligence

IFT — Instruction Fine-Tuning



### Fine Model (RL)

Reinforcement Learning  
alignment step



### Foundation Model

Fully trained & aligned

RLVR

## Pre-Training STAGE 1

Largest compute stage

### Model Architecture (determines parameters)

D / MoE  
Dense / Mixture of Experts



× L Attention (A)  
stacked transformer layers



I/P — Tokenizer  
Input processing

### TRAINING DATA

Crawl / Web English Internet  
Math Code → Multilingual

Scale: Trillions of Tokens  
17B parameter model · trained on 22 trillion tokens

### LONG CONTEXT EXTENSION

Small extra pre-training run at the end.  
Extends context: 4K → 128K tokens  
(Gemini: 1M · DeepSeek: 128K)  
Trains on longer documents so the model can handle longer prompts & outputs.

## Post-Training STAGE 2 & 3

### FOUNDATION DATASETS

Center of every MDLC stage — feeds Pre-Training, SFT, and RL

#### SFT / IFT

Supervised Fine-Tuning /  
Instruction Fine-Tuning

Fine-tune on curated  
table & training data  
Billions of Tokens

#### RL RLVR

Reinforcement Learning  
with Verifiable Rewards

Coarse Correcting  
Thousands → Millions

#### Sharp & Sensitive

→ Contrastive Learning

✓ Code — run + test cases

✓ Math — convert to code

× Free-form text — open problem

### TRAINING DATA FORMAT

{ Q: ... A: ✓ } +1 accepted

{ Q: ... A: ✗ } -1 rejected

→ preference pairs for alignment training

### CRITICALITY PER TOKEN ↑

#### PRE-TRAINING

Low per-token criticality  
Pass anything: Reddit, Twitter, web



#### SFT

Higher sensitivity  
Billions of curated tokens



#### RL

Maximum sensitivity  
One wrong reward = model drifts

## Alignment Algorithms STAGE 3

### DPO

Direct Preference  
Optimization

### GRPO

Group Relative Policy  
Optimization

### GRPO — ENVIRONMENT & ROLLOUTS

Rollouts: { A<sub>1</sub>, A<sub>2</sub> ... A<sub>n</sub> }

#### ENV

Assigned  
Task

Comp Code

Judge / Reward

Code Interpreter

Compiler

Test Cases



>80% pass  
+1 reward

<30% pass  
-1 penalty

1, 2, 3...  
ranked

1 prompt → 16–20 rollouts generated  
Reward model ranks all rollouts  
Model learns from contrastive signal

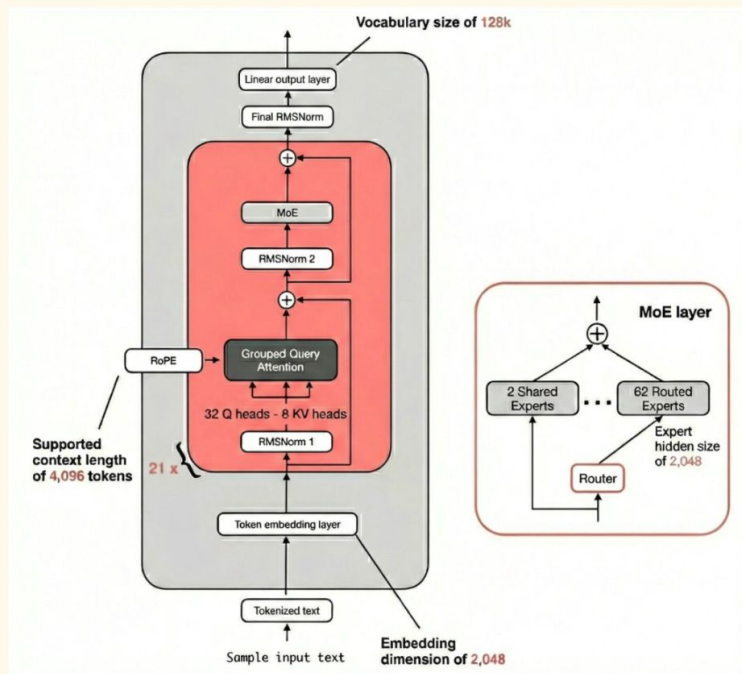
### DPO — PREFERENCE LEARNING (JFT)

Applied to the JFT checkpoint (after SFT).  
1 prompt → 1 chosen + 1 rejected.  
Trains directly on chosen vs. rejected pairs.  
No separate reward model needed.  
→ Contrastive learning signal

# Our Current Flagship Model: Param-17B Mixture of Experts LLM

## Param-2-17B-MOE-A2.4B: High-Efficiency Multilingual Powerhouse

State-of-the-art Mixture-of-Experts (MoE) model with 17B total parameters (2.4B active per token), combining massive knowledge capacity with fast, efficient inference, trained on ~22T tokens across 64 experts. Supports 22 Indian languages with strong code, math, and tool-use reasoning, optimized for culturally grounded, region-specific tasks and inclusive representation.



| Languages supported |             |            |
|---------------------|-------------|------------|
| • Assamese          | • Kannada   | • Punjabi  |
| • Bengali           | • Kashmiri  | • Sanskrit |
| • Bodo              | • Konkani   | • Santali  |
| • Dogri             | • Maithili  | • Sindhi   |
| • Gujarati          | • Malayalam | • Tamil    |
| • Hindi*            | • Manipuri  | • Telugu   |
| • Marathi           | • Nepali    | • Urdu     |
| • Odia (Oriya)      |             |            |

Now available for access through:

- **AIKosh:** [Link](#)
- **Hugging Face:** [Link](#)

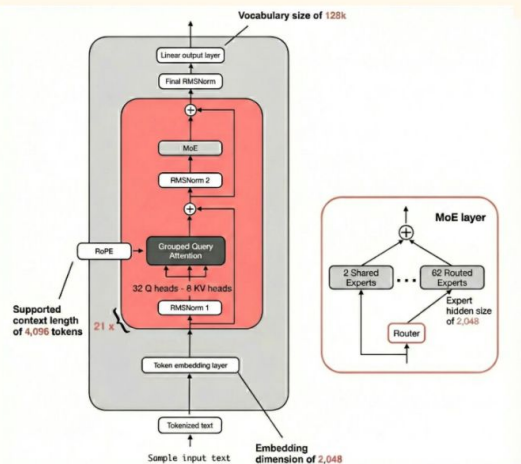
# Our Current Flagship Model: Param-17B Mixture of Experts LLM

## PARAM 2 17B MoE - Hybrid MoE: Efficiency & Specialization

### Architectural Innovation

Employs a unique **Hybrid MoE** approach to solve the "instability" often found in sparse models:

- Shared Experts:** Two experts remain **always active** to maintain stable cross-lingual representations and smooth code-switching.
- Routed Experts:** Six experts dynamically selected per token to handle specialized tasks and specific language nuances.
- Dense Pre-processing:** A dense layer transforms raw inputs into structured representations before MoE, ensuring better training stability.



| Metric           | Value               | Metric              | Value                   |
|------------------|---------------------|---------------------|-------------------------|
| Total Parameters | 17B                 | Active Params/Token | 2.4B                    |
| Total Experts    | 64                  | Top-K Experts       | 6 (Routed) + 2 (Shared) |
| Context Length   | 4096 Tokens         | Vocabulary Size     | 128,000                 |
| Attention        | Grouped Query (GQA) | Precision           | bf16-mixed              |
| Hidden Layers    | 21                  | Activation          | SiLU / RMSNorm          |

**Forward Looking:** A long-context variant supporting up to **128k tokens** currently in development for long-form document processing.

# Training a 17B Model from Scratch

## Massive scale, distributed architecture

17billion-parameter hybrid MoE model

Coordinated distributed training across hundreds of GPUs for millions of steps

Risk of instability, hardware failure, or network bottlenecks

Potential loss of days of training time

## Multilingual data complexity

Training data across 22 Indian languages + English

Required specialised tokenisers and quality classifiers

Careful curation and blending needed

Balance between low-resource and high-resource languages

## Infrastructure resilience

Pre-training runs spanning weeks

Need to handle node failures without stopping training

Requirements: consistent low-latency communication, high bandwidth, efficient checkpointing

## Performance at the frontier

22 trillion tokens to process

Even small per-step inefficiencies compound into days of training time

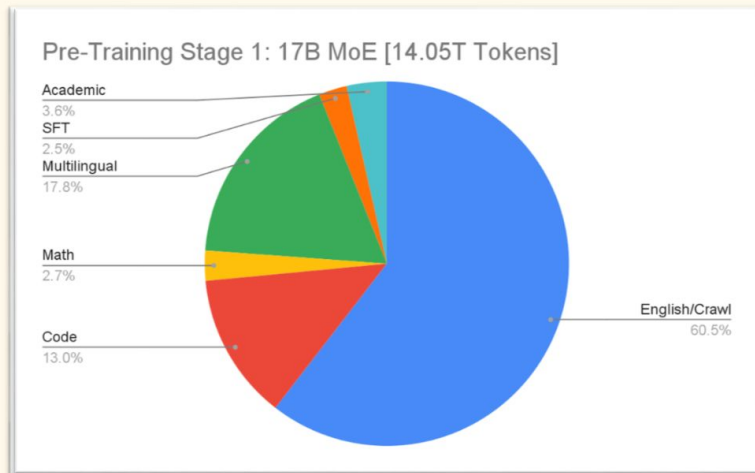
Need to maximise every FLOP from hardware

# Training Dataset details for the Param 17B Foundation Model

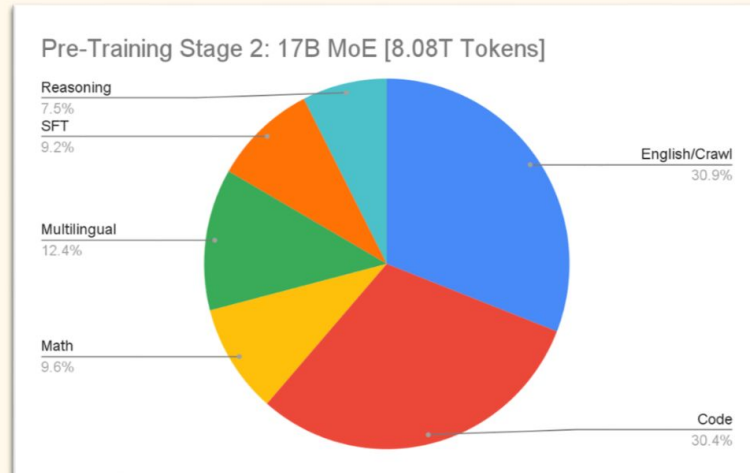
## PARAM 2 17B MoE - DATA DETAILS

Dataset Preparation Details: Multi-Stage Pretraining on 22.13 Trillion Tokens

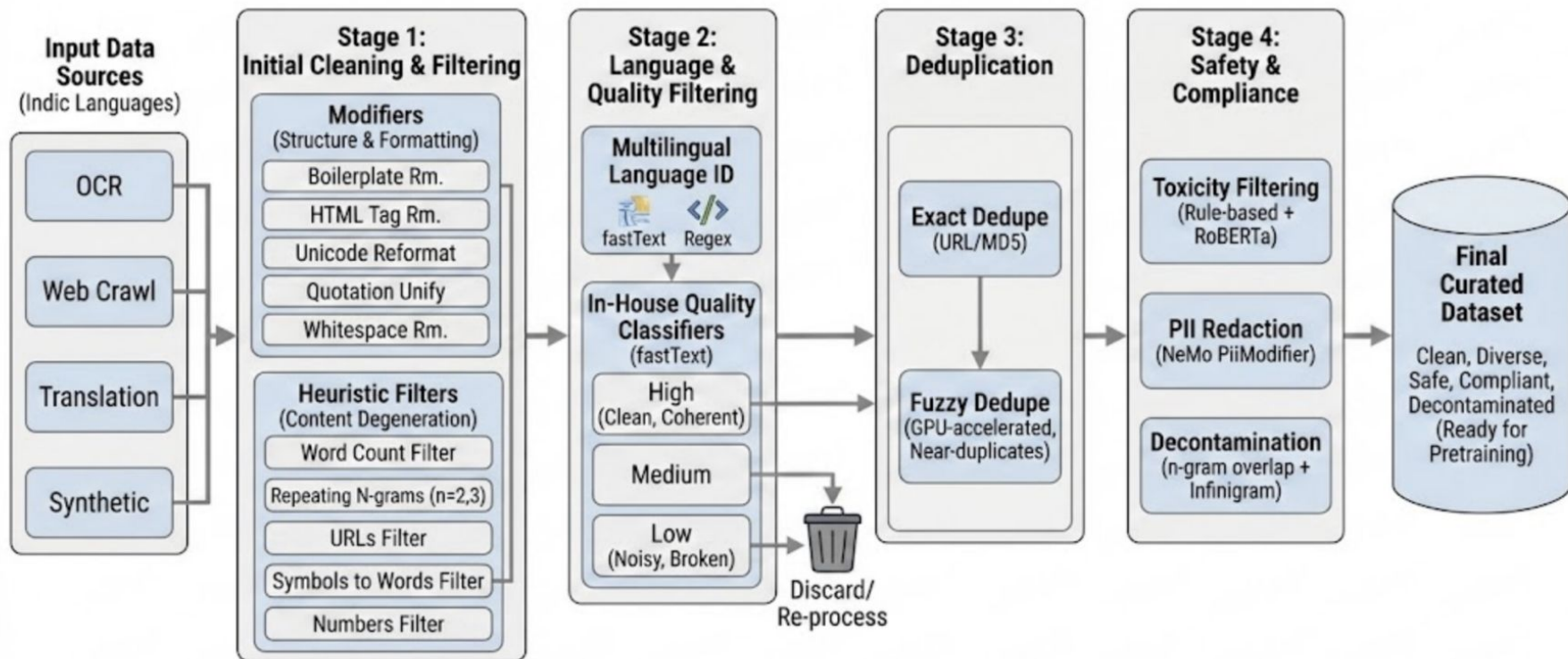
### Pre-Training Stage 1 [14.05T]



### Pre-Training Stage 2 [8.08T]



# Data Curation Process



# Reinforcement Learning for Developing Advanced Capabilities in the LLMs



# Benchmarks and Performance

## PARAM 2-17B MoE – Base Model Performance

**General English Benchmarks:** *Strong foundational performance across standard academic tasks*

| Task          | Param-2-17B-A2.4B | Qwen1.5-MoE-A2.7B | DeepSeek-R1-Distill-Qwen-14B | DeepSeek-V2-Lite-16B | GPT-OSS-20B |
|---------------|-------------------|-------------------|------------------------------|----------------------|-------------|
| ARC Challenge | 56.83             | 44.03             | 53.75                        | 45.31                | 48.98       |
| ARC Easy      | 77.69             | 68.90             | 74.75                        | 70.41                | 75.51       |
| HellaSwag     | 77.43             | 77.26             | 79.04                        | 73.53                | 57.80       |
| SQuADv2       | 33.86             | 13.96             | 23.77                        | 25.20                | 8.66        |
| MMLU          | 57.79             | 62.22             | 73.05                        | 46.57                | 66.37       |

**Code, Math & Reasoning Benchmarks:** *Specialized capabilities for technical problem-solving*

| Task             | Param-2-17B-A2.4B | Qwen1.5-MoE-A2.7B | DeepSeek-R1-Distill-Qwen-14B | DeepSeek-V2-Lite-16B | GPT-OSS-20B |
|------------------|-------------------|-------------------|------------------------------|----------------------|-------------|
| GSM8K (Math)     | 57.32             | 61.49             | 87.87                        | 38.21                | 36.54       |
| MathQA           | 40.23             | 35.61             | 48.44                        | 38.73                | 36.68       |
| HumanEval (Code) | 36.59             | 33.54             | 26.22                        | 26.83                | 24.39       |
| MBPP (Code)      | 47.00             | 38.66             | 61                           | 26.6                 | 72.06       |

# Benchmarks and Performance

## PARAM 2-17B MoE – Base Model Performance

Instruction Following Benchmarks: *Evaluation of comprehension and strict task adherence*

| Task       | Param-2-17B-A2.4B | Qwen1.5-MoE-A2.7B | DeepSeek-R1-Distill-Qwen-14B | DeepSeek-V2-Lite-16B | GPT-OSS-20B |
|------------|-------------------|-------------------|------------------------------|----------------------|-------------|
| TriviaQA   | 54.64             | 38.00             | 17.88                        | 34.62                | 29.05       |
| LogiQA     | 52.35             | 31.18             | 33.49                        | 29.34                | 32.87       |
| WinoGrande | 67.96             | 69.22             | 72.69                        | 67.01                | 67.48       |
| TruthfulQA | 46.42             | 43.57             | 58.63                        | 28.89                | 52.93       |
| IFEval     | 34.05             | 13.19             | 51.80                        | 12.75                | 31.42       |

Indic Language Benchmarks: *Unmatched performance in Indian language understanding and cultural*

| Task                        | Param-2-17B-A2.4B | Qwen1.5-MoE-A2.7B | DeepSeek-R1-Distill-Qwen-14B | DeepSeek-V2-Lite-16B | GPT-OSS-20B |
|-----------------------------|-------------------|-------------------|------------------------------|----------------------|-------------|
| Sanskriti-En                | 66.54             | 28.71             | 74.58                        | 64.84                | 75.14       |
| ARC Challenge (Indic-Multi) | 36.37             | 21.71             | 23.63                        | 21.76                | 31.55       |
| TriviaQA (Indic-Multi MCQ)  | 72.95             | 25.21             | 53.99                        | 29.18                | 71.67       |
| Indic BoolQ                 | 75.98             | 46.48             | 70.03                        | 62.90                | 70.02       |
| HellaSwag Hi                | 49.63             | 25.86             | 39.02                        | 30.23                | 41.25       |
| MMLU Hi                     | 59.23             | 33.27             | 44.52                        | 24.26                | 50.29       |

# Constitutional Morality Index (CMI) — Leaderboard on github

## Combined CMI Leaderboard

| Rank | Model              | CMI  | Caste | Gender Violence | N  |
|------|--------------------|------|-------|-----------------|----|
| 1    | grok-4             | 3.77 | 3.83  | 3.71            | 48 |
| 2    | mistral-medium-3.1 | 3.66 | 3.74  | 3.58            | 47 |
| 3    | mistral-large      | 3.66 | 3.65  | 3.67            | 47 |
| 4    | grok-4-fast        | 3.62 | 3.67  | 3.58            | 48 |
| 5    | deepseek-r1        | 3.47 | 3.42  | 3.52            | 45 |
| 6    | claude-sonnet-4.5  | 3.43 | 3.57  | 3.29            | 47 |
| 7    | grok-3             | 3.38 | 3.33  | 3.42            | 48 |
| 8    | pixtral-large      | 3.33 | 3.25  | 3.42            | 48 |
| 9    | gemini-2.5-flash   | 3.31 | 3.21  | 3.42            | 48 |
| 10   | bharatgen-17b      | 3.31 | 3.46  | 3.17            | 48 |
| 11   | claude-opus-4.1    | 3.29 | 3.33  | 3.25            | 48 |
| 12   | sarvam-m           | 3.29 | 3.33  | 3.25            | 48 |
| 13   | minimax-m2.1       | 3.26 | 3.39  | 3.13            | 46 |
| 14   | gemini-3-flash     | 3.26 | 3.30  | 3.21            | 47 |
| 15   | claude-sonnet-4    | 3.25 | 3.33  | 3.17            | 48 |
| 16   | o4-mini            | 3.25 | 3.12  | 3.38            | 48 |
| 17   | claude-opus-4      | 3.23 | 3.33  | 3.12            | 48 |

## Caste Domain — By Harm Mode

| Rank | Model              | Score | DV   | IV   | CC   | PC   |
|------|--------------------|-------|------|------|------|------|
| 1    | grok-4             | 3.83  | 4.00 | 3.83 | 4.00 | 3.50 |
| 2    | mistral-medium-3.1 | 3.74  | 4.00 | 3.83 | 3.80 | 3.33 |
| 3    | grok-4-fast        | 3.67  | 3.83 | 3.50 | 3.83 | 3.50 |
| 4    | mistral-large      | 3.65  | 4.00 | 3.50 | 3.83 | 3.20 |
| 5    | claude-sonnet-4.5  | 3.57  | 3.83 | 3.67 | 3.67 | 3.00 |
| 6    | bharatgen-17b      | 3.46  | 4.00 | 3.33 | 3.50 | 3.00 |
| 7    | deepseek-r1        | 3.42  | 4.00 | 3.17 | 3.50 | 3.00 |
| 8    | minimax-m2.1       | 3.39  | 4.00 | 3.50 | 3.00 | 3.00 |
| 9    | sarvam-105b        | 3.38  | 3.67 | 3.67 | 3.50 | 2.67 |
| 10   | claude-opus-4.1    | 3.33  | 3.67 | 3.50 | 3.50 | 2.67 |
| 11   | claude-opus-4      | 3.33  | 4.00 | 3.33 | 3.50 | 2.50 |
| 12   | claude-sonnet-4    | 3.33  | 4.00 | 3.50 | 3.50 | 2.33 |
| 13   | grok-3             | 3.33  | 3.83 | 3.33 | 3.67 | 2.50 |
| 14   | sarvam-m           | 3.33  | 3.67 | 3.50 | 3.67 | 2.50 |



# **BharatGen's Sovereign Multimodal Models**

## Shrutam2: Speech to Text Model

- **Multilingual ASR across 12 Indic languages:** Hindi, Marathi, Tamil, Telugu, Malayalam, Kannada, Odia, Bengali, Urdu, Assamese, Gujarati, Punjabi.
- Uses Conformer encoder, Projector and our own **Param-1B LLM** for speech recognition
- **Dynamic Multi-Expert Projectors** enabling the model to unify learning across languages while adapting to their unique characteristics.

Access model at

- Hugging face: [Shrutam2](#)
- AI Kosh: [Shrutam2](#)
- Demo link: <https://voices.bharatgen.dev/shrutam/>



# Shrutam2: Speech to Text Model

- We compare the performance of Shrutam2 with IndicWhisper on **Vistaar and IndicVoices** Benchmark
- We use the **WER** (Word error rate) to compare the performance as shown in the table, where lower WER is better performance.

| Language  | Shrutum2     | IndicWhisper |
|-----------|--------------|--------------|
| Bengali   | <b>14.59</b> | 28.52        |
| Gujarati  | <b>15.30</b> | 26.85        |
| Hindi     | <b>14.08</b> | 15.96        |
| Kannada   | <b>17.33</b> | 31.12        |
| Malayalam | <b>27.44</b> | 42.49        |
| Marathi   | <b>16.08</b> | 23.68        |
| Odia      | <b>24.29</b> | 35.47        |
| Punjabi   | <b>13.93</b> | 26.34        |
| Assamese  | <b>23.15</b> | NA           |
| Tamil     | <b>24.70</b> | 32.37        |
| Telugu    | <b>21.48</b> | 35.19        |
| Urdu      | <b>12.61</b> | 24.72        |



# Addressing the Indian Document Challenge



## 22+ Languages

Vast linguistic diversity creating barriers to unified data processing.



## Complex Layouts

Multi-page, non-standard formats in official and legal documents.



## Healthcare Records

Critical need for accurate digitization of handwritten and printed medical data.



## Legal Complexity

Regional variations and intricate legal jargon challenge generic models.

Foreign VLMs **fail** on Indian scripts, regional variations, and official document formats.

## Usecase Sectors

Governance

Finance

Healthcare

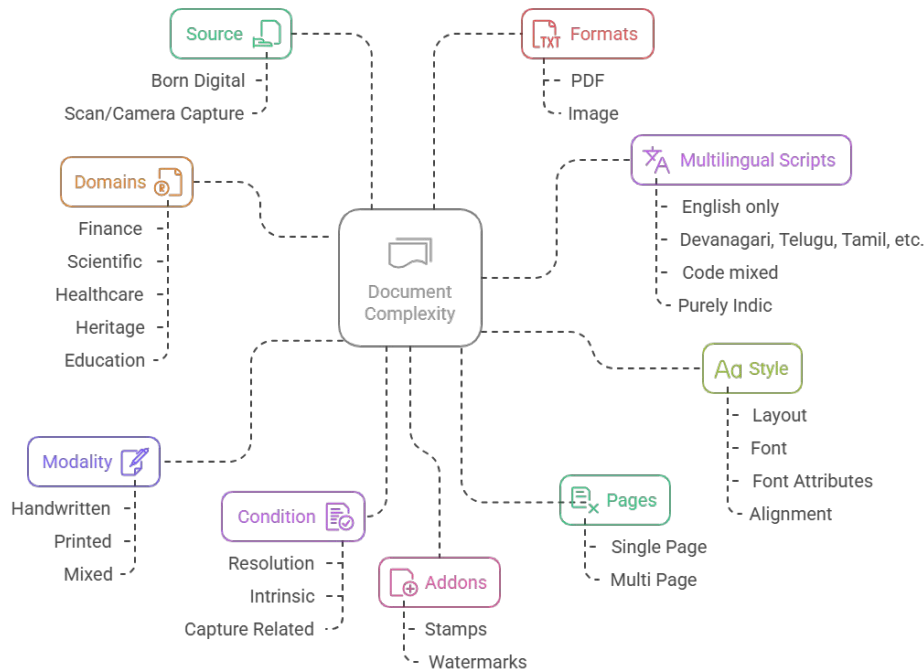
Legal

Defense

Agriculture

Education

Public Service



# India-centric Vision Foundation Model from Scratch



## Multi-page Layout Understanding

Comprehends complex structures across multiple pages, essential for reports and contracts.



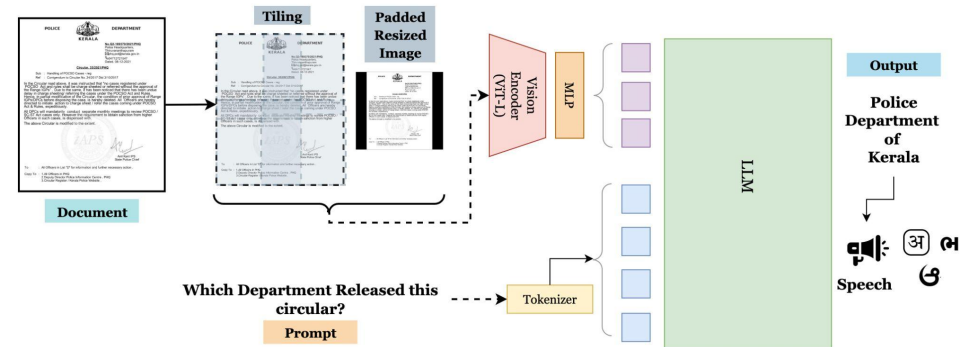
## End-to-End Models

Integrated pipeline for OCR, VQA, Classification, and Summarization.



## Multilingual & Domain-Specific

Fine-tuned for Finance, Legal, Healthcare, and Governance in 22 Indian languages.



# Use Case: In-house built Ayurveda Specific Foundation Model



## BhashaBench - Ayur

First comprehensive benchmark evaluate models on traditional Ayurvedic knowledge and practice

English & Hindi

15+ specialized Ayurvedic disciplines

Traditional knowledge for clinical use, herbal pharmacology, and holistic healthcare

Multiple Choice Questions (MCQ), Fill in the Blanks, Match the Column, Assertion-Reasoning

### Models Evaluated:

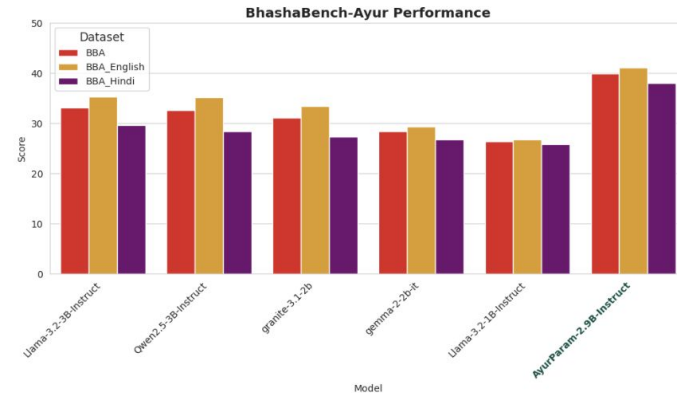
29 (Qwen3-235B, Gemma-2-27B, Llama-3.1-8B)

### Top Accuracy:

English: 60.25%  
Hindi: 54.78%

## Ayur Param

- Fine-tuned on 1000+ Ayurvedic texts (~54M words).
- Bilingual (English + Hindi), context-aware responses grounded in Ayurveda.
- Handles Ayurvedic queries classical text interpretation, clinical guidance, and wellness knowledge.
- Supports multiple query types – Q&A, reasoning, MCQ, and clinical use cases.





**Genomics Use Case being built on top of BharatGen's Param Model**



# Training Curriculum & Data Specification

Fine-Tuning PARAM 2 (17B MoE) for Genomic Understanding  
via Natural Language Convergence

---

# The Convergence Hypothesis

## Nucleotide-Token Models

(Evo2, DNABERT, Nucleotide Transformer)

- Tokenize DNA as A/C/G/T k-mers
- Learn statistical sequence patterns
- Strong on coding regions
- Weak on regulatory context
- No clinical reasoning

## Text-Based Approach (Ours)

(PARAM 2 + Genomic Fine-Tuning)

- Reason about genomic concepts in NL
- Encode regulatory annotations
- Condition on cell type & disease
- Leverage clinical literature
- Complement sequence models



Knowledge-level convergence, not tokenizer-level. PARAM 2's 22T token pre-training (incl. medical literature) provides a warm start. We extend to molecular-level genomic reasoning.

# Four-Phase Training Curriculum

Increasing biological complexity, mirroring your paper's 4-stage evaluation

1

**Genomic Knowledge  
Foundation**

**25-30B tokens**

3-4 weeks

Gene function, molecular  
biology, pathway reasoning

2

**Variant Interpretation  
& Clinical Genomics**

**30-35B tokens**

4-5 weeks

ClinVar, ACMG reasoning,  
trait-stratified VEP

3

**Regulatory &  
Epigenomic Context**

**15-20B tokens**

3-4 weeks

ENCODE, FANTOM5,  
cell-type conditioning

4

**Multi-Omics  
Integration**

**15-25B tokens**

3-4 weeks

GTEx, proteomics,  
cross-modal reasoning

**Total: ~85-110 Billion Tokens | ~3-4 Months | 64x H100 GPUs**

# 1

# Genomic Knowledge Foundation

25-30B tokens | 3-4 weeks | CPT + SFT

| Dataset                       | Content                             | Est. Tokens |
|-------------------------------|-------------------------------------|-------------|
| PubMed Central OA (genomics)  | Full-text research papers (~800K)   | ~12B        |
| UniProt/Swiss-Prot            | Protein function annotations (570K) | ~2B         |
| Gene Ontology + KEGG/Reactome | Biological processes & pathways     | ~1.3B       |
| OMIM + GeneReviews            | Gene-disease descriptions           | ~1.2B       |
| NCBI Gene + Organism DBs      | Gene summaries, cross-species       | ~2B         |
| Synthetic Q&A Pairs           | Genomics reasoning (5M pairs)       | ~6B         |
| Open-Access Textbooks         | Molecular bio & genetics            | ~500M       |

## Evaluation Checkpoints

- Gene function retrieval: Can it describe BRCA1, TP53 with expert-level detail?
- Pathway reasoning: Given a gene, enumerate relevant pathways and interaction partners
- Cross-species homology awareness: Conserved vs. divergent gene functions

## 2

# Variant Interpretation & Clinical Genomics

30-35B tokens | 4-5 weeks | SFT + Chain-of-Thought

| Dataset                       | Content                              | Tokens |
|-------------------------------|--------------------------------------|--------|
| ClinVar (3.6M variants)       | Classifications + review text        | ~4B    |
| gnomAD v4.x (800M variants)   | Population freq. & constraint        | ~5B    |
| CADD / REVEL / AlphaMissense  | Computational pathogenicity (190M)   | ~8B    |
| TCGA (33 cancer types)        | Somatic mutations + clinical annot.  | ~3B    |
| Synthetic clinical cases (2M) | ACMG reasoning chains                | ~6B    |
| TraitGym + IDH glioma cohort  | Trait-variant benchmarks             | ~700M  |
| PharmGKB + HGMD/LOVD          | Pharmacogenomics + disease mutations | ~2.5B  |

## Critical Design Decisions

### Trait-Stratified Training

Balanced by biological category  
(developmental, immune, metabolic,  
neurological, oncological, structural)

### ACMG Reasoning Chains

Train WHY variants are pathogenic, not just  
classify. Chain-of-thought with evidence  
criteria.

### Population Context

gnomAD→natural language. Critical for  
India-specific clinical genomics  
interpretation.

## 3

# Regulatory & Epigenomic Context

15-20B tokens | 3-4 weeks | CPT + Context-Conditioned SFT

| Dataset                              | Tokens |
|--------------------------------------|--------|
| ENCODE Project (1.6M elements)       | ~3B    |
| Roadmap Epigenomics (127 tissues)    | ~2B    |
| FANTOM5 enhancer atlas (65K)         | ~1B    |
| 4DN Hi-C / chromatin 3D              | ~1.5B  |
| GEO expression datasets (50K)        | ~3B    |
| lncRNA CRISPR screens (5 cell lines) | ~500M  |
| RNAcentral + Rfam                    | ~2B    |
| Synthetic regulatory reasoning       | ~1.5B  |

## Context-Conditioning Innovation

Each data point augmented with explicit context:

*"In [K562 / bone marrow / CML], the enhancer at chr8:128M shows [ATAC: open / H3K27ac: high / Hi-C: MYC contact]. Active here but inactive in [HepG2 / liver]."*

Mirrors your finding that cellular context dominates predictive variability.

## Multi-Species Regulatory Coverage

*E. coli*

Prokaryotic  
regulatory nets

*S. cerevisiae*

Yeast  
transcription

*C. elegans*

Developmental  
regulation

*D. melanogaster*

Enhancer-  
promoter logic

*M. musculus*

Mammalian  
epigenomics

*H. sapiens*

Comprehensive  
human regulation

## 4

# Multi-Omics Integration & Reasoning

15-25B tokens | 3-4 weeks | SFT + Chain-of-Thought



## OpenGenome2 Metadata

Taxonomic & functional annotations from the 9T bp dataset used to train Evo2

~3B



## GTEx (54 tissues)

Tissue-specific expression profiles and eQTL data across 17K samples

~2B



## Human Protein Atlas

Protein expression & subcellular localization for 20K proteins

~1B



## STRING Interactions

12M protein-protein interactions converted to graph descriptions

~2B



## HMDB Metabolomics

220K metabolite-gene-disease linkages

~1B



## Synthetic Cross-Modal

"Given variant X in gene Y, predict expression, protein, pathway effects"

~6B

**Thank You**